



1. Simplified form of the logic expression $(A + \bar{B} + C)(A + \bar{B} + \bar{C})(A + B + C)$

A. $\bar{A}B + \bar{C}$ B. $A + \bar{B}C$ C. A D. $AB + \bar{C}$

2. The minimal POS function $G(A,B,C,D)$ which represent by following K-map is?

	CD	00	01	11	10
AB					
00	0	1	1	0	
01	1	1	1	1	
11	1	1	0	0	
10	0	0	0	0	

A. $G(A,B,C,D) = (\bar{A} + B)(B + D)(\bar{A} + \bar{C})$
 B. $G(A,B,C,D) = \bar{A}.D + B.\bar{C} + \bar{A}.B$
 C. $G(A,B,C,D) = \prod M(0,2,8,9,10,11,14,15)$
 D. $G(A,B,C,D) = (A + \bar{B})(\bar{B} + \bar{D})(A + C)$

3. Binary data is being represented in size of byte and in 2's complement form. The number of 0's in the representation of $\{-127\}$ decimal is

A. 8 B. 7 C. 6 D. 5

4. For a binary half-subtractor having two inputs A and B, the correct set of logical expressions for the outputs D (= A minus B) and X (= borrow) are?

A. $D = AB + A'B, X = A'B$ B. $D = A'B + AB' + AB, X = AB'$ C. $D = A'B + AB', X = A'B$ D. $D = AB + A'B', X = AB'$

5. The minterm expansion of $f(P, Q, R) = PQ + QR + PR$ is

A. $m_2 + m_4 + m_6 + m_7$ B. $m_0 + m_1 + m_3 + m_5$ C. $m_2 + m_3 + m_4 + m_5$ D. $m_0 + m_1 + m_6 + m_7$

6. Consider minterm expansion of the function $F(P, Q, R, S) = \Sigma(0, 5, 10, 15)$ and $F(P, Q, R, S)_{\text{DON'T CARE}} = \Sigma(2, 7, 8, 13)$ The minimal sum of product form of F is

A. $Q\bar{S} + \bar{Q}S$ B. $\bar{Q}RS + Q\bar{R}S + QR\bar{S} + \bar{Q}\bar{R}S$
 C. $\bar{Q}\bar{S} + QS$ D. $PQS + P\bar{Q}S + P\bar{Q}\bar{S} + P\bar{Q}S$

7. The function $F =$

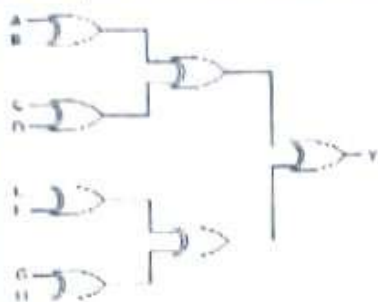
A. $X\bar{Y}Z + \bar{X}YZ$ B. $\bar{X}YZ + \bar{X}Y\bar{Z}$ C. $\bar{X}Y\bar{Z} + XYZ$ D. $X\bar{Y}\bar{Z} + \bar{X}Y\bar{Z}$



8. Minimized expression for $(x + y) + (\overline{x + y}) + (\overline{xy}) + \overline{x}$ is

A. x B. xy C. y D. $x + y$

9. The 8-input XOR circuit shown has an output of $Y = 1$. Which input combination below (ordered A – H) is correct?



A. 10111100

B. 101,11000

C. 11100111

00011101

10. 01101000_{BCD}

+ 00110110_{BCD}

= _____ BCD

A. 10011110

B. 100000100

C. 000100000100

001000001000

11. The following switching functions are to be implemented using a decoder

1. $f_1 = \sum m(1, 2, 4, 8, 10, 14)$

2. $f_2 = \sum m(2, 5, 9, 15)$

3. $f_3 = \sum m(2, 4, 5, 6, 7)$

The minimum configuration of decoder is

A. 2 to 4 line

B. 3 to 8 line

C. 4 to 16 line

D. 5 to 32 line

12. For the K map of the given figure, the simplified Boolean expression is

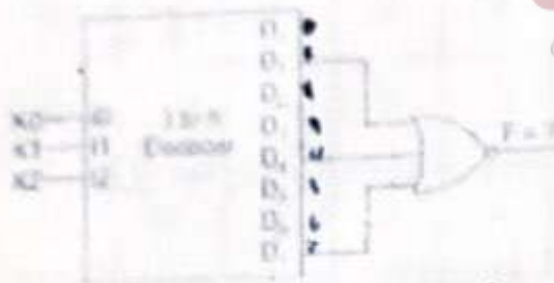
CD \ AB	C			
	00	01	11	10
00	1	1		1
01	1	1		1
11			1	1
10				

A. $A'C' + A'D' + ABC$ B. $AB'D + BC$ C. $AC'D + AC$ D. $AC'D + AC + BC$



13. 5 to 32 decoder can be implemented by using -----
- ☒ A. one 2 to 4 decoder and four 3 to 8 decoders
 - B. four 3 to 8 decoders
 - C. three 3 to 8 decoders
 - D. eight 2 to 4 decoders

14. A 3 to 8 decoder is shown in the figure below find the function $F(x_2, x_1, x_0)$ -----



- A. $\Sigma m(1,4,7)$ ☒ B. $\Sigma m(0,2,3,5,6)$ C. $\Sigma m(0,2,3,5,7)$ D. $\Sigma m(1,2,4,7)$
15. Two 4-bit binary numbers (1011 and 1111) are applied to a 4-bit parallel adder. The carry input is 1. What are the values for the sum and carry output?
- A. $S_3 S_2 S_1 S_0 = 0111$, Cout = 0
 - B. $S_3 S_2 S_1 S_0 = 1111$, Cout = 1
 - ☒ C. $S_3 S_2 S_1 S_0 = 1011$, Cout = 1
 - D. $S_3 S_2 S_1 S_0 = 1100$, Cout = 1